

Balancing Efficiency and Fairness in the Integrated Truck–Drone Dispatching Problem with Dynamic Endurance via Pareto Front Grid Guided Multi-objective Optimization

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Abstract. The coordination of trucks and drones in logistics systems offers new opportunities for cost-effective and sustainable logistics operations while enhancing customer satisfaction, yet significant challenges arise from heterogeneous vehicle capabilities and the payload-dependent endurance of drones. This paper introduces Integrated Truck–Drone Dispatching Problem with Dynamic Endurance (ITDDPDE), a novel multi-objective formulation that jointly schedules a fleet of trucks and drones, capturing two realistic features of drone operations: (i) dynamic endurance that depends on the payload carried, and (ii) flexible launch and retrieval points along truck routes. The problem simultaneously minimizes three objectives: total operational cost, fairness across customers, and fairness among vehicles. To tackle the ITDDPDE problem, we propose a multi-objective evolutionary algorithm based on Pareto front grid decomposition, incorporating two-level encoding and a heuristic mechanism to determine drone launch and landing points. Numerical experiments on diverse benchmark scenarios demonstrate that the proposed algorithm significantly outperforms state-of-the-art multi-objective algorithms in terms of Hypervolume and Inverted Generational Distance. The source code is available at: <https://github.com/BuiXuanSon1412/PFG-TS.git>.

Keywords: vehicle routing problem, multi-objective optimization, fairness, evolutionary computation, Pareto front grid

1 Introduction

The rapid development of e-commerce and on-demand logistics has driven the need for innovative transportation systems that can meet customer expectations while maintaining operational efficiency [13]. Traditional truck-based transportation systems often face limitations in urban areas due to traffic congestion, high operating costs, and environmental concerns [27]. To address these issues, drones have been increasingly deployed as a complementary transportation technology thanks to their speed, flexibility, and ability to access hard-to-reach locations [10,26], demonstrating their potential to reduce operating costs and improve service quality.

However, several practical challenges remain largely underexplored. First, drones are typically modeled with static flight endurance, ignoring the critical dependence of energy consumption on payload weight [14,4]. Moreover, many models assume that each drone can only launch from and land at the depot or a single truck, or that a drone serves only one customer per trip. Such assumptions restrict operational flexibility and significantly reduce drone efficiency, as drones in practice can launch from a truck and land at any location, serving multiple customers per trip. Second, while system efficiency has been the primary focus of optimization models, fairness among vehicles and fairness among customers, which is an essential factor for sustainability and long-term operability, has received limited attention [24]. To bridge these gaps, we introduce the ITDDPDE problem, which jointly schedules a homogeneous fleet of trucks and drones with the objectives of minimizing operational costs and promoting fairness through balanced workload allocation across vehicles and equitable service to customers. To effectively solve ITDDPDE, then, we propose a novel algorithm, called Pareto Front Grid guided Multi-objective Evolutionary Algorithm with Time Synchronization (PFG-TS). Our main contributions are summarized as follows:

- We introduce the ITDDPDE problem, incorporating (i) payload-dependent drone endurance and (ii) flexible launch and retrieval points along truck routes. The problem simultaneously optimizes three objectives: total operational cost, fairness among customers, and fairness across vehicles.
- We propose a Pareto-based evolutionary optimization framework with a problem-specific two-level encoding and a heuristic mechanism to determine flexible drone launch and landing points, ensuring coordinated scheduling of trucks and customer visits.
- Numerical experiments are conducted across various benchmark instances, demonstrating significant improvements over state-of-the-art baselines in terms of Hypervolume (HV) and Inverted Generational Distance (IGD). Additionally, we analyze the factors influencing algorithm performance, highlighting the efficiency gains of integrated truck-drone systems compared to traditional truck-only models.

2 Related works

With the growing interest in hybrid truck–drone delivery systems, prior research has primarily focused on synchronized vehicle routing models, where trucks wait for drone launch and retrieval [1]. More recent approaches have introduced parallel models, allowing trucks and drones to operate with greater independence and thereby improving overall efficiency [19,21]. Another distinction concerns the number of drone trips: while initial studies assumed single-trip drones carrying one package and returning immediately to the truck, current models permit multi-trip drones that can serve multiple customers before rejoining, exploiting their endurance more effectively [20]. Operational analyses have further compared stationary truck modes, moving truck modes, and hybrid combinations, consistently showing that hybrid coordination achieves the best trade-off between delivery efficiency and system complexity. Beyond efficiency, fairness

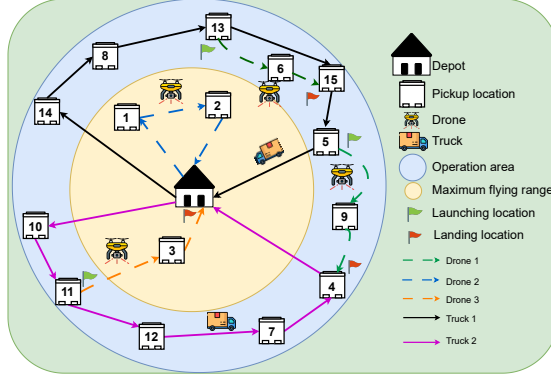


Fig. 1: An example of ITDDPDE solution.

has increasingly been recognized as an important consideration in routing and scheduling [12,9]. Many applications require balancing workloads among vehicles and drivers to ensure equity, sustainability, and operational robustness [16]. Additionally, drone endurance has been modeled with increasing realism by incorporating payload-dependent energy consumption [18], providing more reliable and implementable routing decisions compared to static range assumptions.

From a methodological perspective, exact methods such as mixed-integer linear programming and dynamic programming have been applied to various VRP variants, but their scalability is limited as problem size grows, making them unsuitable for complex truck-drone systems [3]. This has motivated the use of metaheuristics, with evolutionary algorithms (EAs) emerging as versatile approaches for solving combinatorial optimization problems [6,7,15]. Their population-based nature allows broad exploration of high-dimensional search spaces, while variation operators promote diversification and prevent premature convergence [5,8]. Building on this foundation, multi-objective EAs (MOEAs) extend EAs to scenarios with multiple conflicting objectives, maintaining diverse solutions along the Pareto front rather than collapsing them into a single outcome [22]. Such flexibility has enabled MOEAs to achieve notable success in logistics, supply chains, and related domains [23,12].

Building on these insights, this paper introduces the ITDDPDE problem, which integrates (i) payload-dependent drone endurance and (ii) flexible launch and retrieval points along truck routes into a unified hybrid truck-drone scheduling framework. The problem simultaneously optimizes three objectives: total operational cost, fairness among customers, and fairness across vehicles. To address its computational complexity, we propose a Pareto-based evolutionary optimization framework with problem-specific two-level encoding and a heuristic mechanism to determine flexible drone launch and landing points, ensuring coordinated scheduling of trucks and customer visits.

3 Problem formulation

The operational environment of a typical ITDDPDE is represented as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, as in Fig. 1. Specifically, $\mathcal{V} = \{s\} \cup \mathcal{N}$, where s denotes the depot

Table 1: Decision variables

Variables	Description
R	Set of drone trips
x_i^k	1 if customer i is served by truck k , 0 otherwise
$\tilde{x}_i^{dr}$	1 if customer i is served by drone d on route r , 0 otherwise
y_{ij}^k	1 if truck k travels directly from node i to j , 0 otherwise
w^k	1 if truck k activate, 0 otherwise
$\tilde{w}^d$	1 if drone d activate, 0 otherwise
$\tilde{y}_{ij}^{dr}$	1 if drone trip r flies from node i to j , 0 otherwise
$\tilde{a}_i$	Time when customer i is served
ξ_{ij}^{dr}	Payload carried by drone d along arc (i, j) on route r
W^k	total quantity of goods received by truck k from all drones.

(equivalent to the provider's warehouse receiving goods), and $\mathcal{N} = \{1, 2, \dots, N\}$ is the set of n pickup customer locations. The edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ consists of arcs connecting distinct locations. A directed edge from node i to j , denoted $(i, j) \in \mathcal{E}$, exists if a feasible travel path is available between i and j . Let d_{ij} and $\tilde{d}_{ij}$ denote the travel distances from i to j for trucks and drones, respectively. For practical purposes, truck distances are measured using the Manhattan metric, while drone distances are computed with the Euclidean metric [17]. The system operates a fleet of K homogeneous trucks, denoted $\mathcal{K} = \{1, 2, \dots, K\}$. Each truck departs from the depot, visits an assigned subset of pickup customers to collect goods, and returns to the depot. Every truck has a maximum capacity Q , representing the upper limit of goods that the truck can carry at any time. To improve system efficiency, a fleet of D homogeneous drones is also deployed, denoted $\mathcal{D} = \{1, 2, \dots, D\}$. Each drone has a maximum payload $\tilde{Q}$, which is considerably smaller than that of a truck. The travel speeds of trucks and drones are represented by v and $\tilde{v}$, respectively. Furthermore, each drone is equipped with a limited battery capacity E , where the payload weight influences the rate of energy consumption during operation. Trucks and drones operate independently, with drones making multiple trips and replacing batteries at the depot or on a truck. Drones can collect goods directly from customers and rendezvous with trucks or the depot for transfer. At each rendezvous, vehicles wait for each other to enable goods transfer and battery replacement, during which drones require τ_l and τ_r for launching and landing. Each request $i \in \mathcal{N}$ is characterized by a tuple $\langle u_i, a_i, t_i \rangle$, where u_i is the amount of goods to be collected at customer i , a_i is the arrival time at which the goods become available for pickup, and t_i is the on-site service time required to load the goods onto a vehicle. In summary, decision variables are listed in Table 1.

This paper adopts the payload–duration-based model proposed in [18] to evaluate the energy consumption of drones. For a given route r performed by drone d , the energy consumed when flying over arc (i, j) is defined as

$$\tilde{E}_{ij}^{dr} = \lambda(\tilde{q}_0 + \xi_{ij}^{dr}) \left(\frac{\tilde{d}_{ij}}{\tilde{v}} + \tau_l \right), \quad (1)$$

where λ denotes the energy consumption rate per unit weight per unit time, $\tilde{q}_0$ is the no-load weight of a drone, and ξ_{ij}^{dr} represents the payload carried by drone d along arc (i, j) . Furthermore, the energy consumed at customer j is given by

$$\hat{E}_{ij}^{dr} = \lambda(\tilde{q}_0 + \xi_{ij}^{dr})\tau_r + \lambda(\tilde{q}_0 + \xi_{ij}^{dr} + u_j)\tilde{x}_j^{dr}t_j + \lambda(\tilde{q}_0 + \xi_{ij}^{dr})h_j^{dr}, \quad (2)$$

where the first term represents the energy consumed during landing. The second term corresponds to the energy consumed for servicing customer j ; if j serves only as a rendezvous point where the drone lands on a truck, this value becomes zero. Finally, the third term captures the energy consumed while hovering at node j , where h_j^{dr} denotes the hovering duration of drone d at customer j on route r . This duration accounts for two cases: the waiting time until the goods at customer j become available if j is served by the drone, or the waiting time until the truck arrives if j is a rendezvous location.

Total Cost. To enhance the overall efficiency of the system, the routing and scheduling of both trucks and drones should also aim at minimizing the total operational cost. This cost reflects the distance-based expenses of all vehicles and is formulated as

$$f_1 = \sum_{k \in \mathcal{K}} \left(\sum_{\substack{i,j \in \mathcal{N} \\ i \neq j}} C^T y_{ij}^k d_{ij} + w^k C_{fix}^T \right) + \sum_{d \in \mathcal{D}} \left(\sum_{r \in \mathcal{R}} \sum_{\substack{i,j \in \mathcal{N} \\ i \neq j}} C^D y_{ij}^{dr} \tilde{d}_{ij} + \tilde{w}^d C_{fix}^D \right), \quad (3)$$

where C^T and C^D denote the per-unit-distance travel costs of trucks and drones, respectively. C_{fix}^T and C_{fix}^D denote the fixed activation costs for a single truck and a single drone, respectively.

Customer-based Fairness. If a vehicle arrives at a location before the order is ready, it waits until the task can be performed; if it arrives after the order is ready, the customer experiences a delay, reducing service quality. To capture fairness from the customer perspective, we consider the maximum waiting time for all customers

$$f_2 = \max_{i \in \mathcal{N}} \{ \max(0, \tilde{a}_i - a_i) \}, \quad (4)$$

Vehicle-based Fairness. To ensure balanced utilization of resources, the fairness of workload distribution among vehicles is explicitly considered. The workload of each vehicle is measured by the total distance traveled. Rather than focusing exclusively on minimizing operational cost, the objective also emphasizes preventing any single vehicle from being disproportionately overloaded. Fairness is quantified by the deviation of individual vehicle workloads from the corresponding average workload within each vehicle category.

$$f_3 = \sqrt{\frac{1}{K} \sum_{k \in \mathcal{K}} (L_k^T - \bar{L}^T)^2} + \sqrt{\frac{1}{D} \sum_{d \in \mathcal{D}} (L_d^D - \bar{L}^D)^2}, \quad (5)$$

where, $L_k^T = \sum_{i \neq j} y_{ij}^k d_{ij}$ denotes total travel distance of truck k , with $\bar{L}^T = \frac{1}{K} \sum_{k \in \mathcal{K}} L_k^T$ as average over all trucks. Similarly, $L_d^D = \sum_{r \in \mathcal{R}} \sum_{i \neq j} y_{ij}^{dr} \tilde{d}_{ij}$ is total flight distance of drone d , and $\bar{L}^D = \frac{1}{D} \sum_{d \in \mathcal{D}} L_d^D$ is average over all drones.

Algorithm 1 PFG-TS algorithm

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1: Generation  $G \leftarrow 0$ ;
2: Initialize population  $\mathbf{P}^G = \{\mathbf{x}^{i,G}\}_{i=1}^{pop\_size}$  with truck-assigned probability  $\alpha$ ;
3: Repair and evaluate  $\mathbf{x}^{i,G}, \forall \mathbf{x}^{i,G} \in \mathbf{P}^{i,G}$ ;
4: while not termination do
5:   Generate Pareto front grid  $PFG$  based on  $\mathbf{P}^G$ ;
6:   Generate offspring population  $\mathbf{O}^{i,G}$  using PMX crossover and bit-flip mutation;
7:   Repair and evaluate  $\mathbf{o}^{i,G}, \forall \mathbf{o}^{i,G} \in \mathbf{O}^{i,G}$ ;
8:    $\mathbf{P}^G = \text{select}(\mathbf{P}^G \cup \mathbf{O}^G)$ ;  $\triangleright$  use non-dominated sorting and crowding-distance
9: end while
10: return Non-dominated Pareto front

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In summary, the ITDDPDE problem is formulated as a MOO model:

$$\text{minimize } \{f_1, f_2, f_3\}, \quad (6)$$

$$\text{subject to } \sum_{k \in \mathcal{K}} x_i^k + \sum_{d \in \mathcal{D}} \sum_{r \in \mathcal{R}} \tilde{x}_i^{dr} = 1, \quad \forall i \in \mathcal{N}, \quad (7)$$

$$\sum_{i \in \mathcal{N}} x_i^k u_i + W^k \leq Q, \quad \forall k \in \mathcal{K}, \quad (8)$$

$$\xi_{ij}^{dr} \leq \tilde{Q}, \quad \forall (i, j) \in \mathcal{E}, d \in \mathcal{D}, r \in \mathcal{R}, \quad (9)$$

$$\sum_{(i,j) \in \mathcal{E}} (\tilde{E}_{ij}^{dr} + \hat{E}_{ij}^{dr}) \leq E, \quad \forall d \in \mathcal{D}, r \in \mathcal{R}, \quad (10)$$

$$\tilde{a}_i \geq a_i, \quad \forall i \in \mathcal{N}, \quad (11)$$

Constraint (7) ensures that each customer is served exactly once, either by a truck or by a drone. Constraints (8) and (9) guarantee that the vehicle payload does not exceed capacity. Constraint (10) restricts drone operations to within their endurance limits. Finally, constraint (11) ensures that all service operations respect customer availability times. In addition, trucks and drones are required to return to the depot at the end of their operations, and drones are not allowed to run out of energy while in flight.

4 Proposed algorithm

To tackle the problem, we propose a novel algorithm, called PFG-TS, that optimizes multiple objectives based on the Pareto front decomposition for reproduction [9]. This paper utilizes a two-level encoding to represent solutions, where the lower level specifies the assignment of orders to either drones or trucks, while the upper level determines the set of orders allocated to each vehicle and the visiting sequence of customers. Detailed algorithm is illustrated in Algorithm 1.

4.1 Individual representation and population initialization

Each individual is represented by a two-level encoding. At the upper level, a permutation of integers from 1 to $N + K + D - 2$ is used: values 1 to N represent

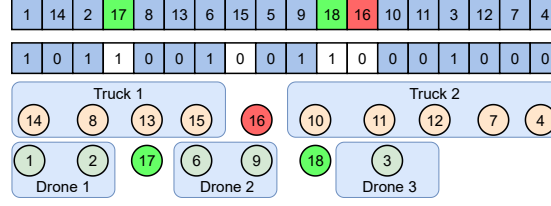


Fig. 2: Individual representation and customer assignment.

customers, $N + 1$ to $N + K - 1$ serve as separators for trucks, and $N + K + 1$ to $N + K + D - 1$ as separators for drones. Fig. 2 illustrates an example with 15 customers, where 16, 17, and 18 act as separators. At the lower level, a binary vector indicates assignment, with 0 for trucks and 1 for drones. Individuals are initialized by random upper-level permutations and probabilistic assignments: each customer is assigned to a truck with probability α and to a drone with probability $1 - \alpha$. Since drones usually have a smaller workload compared to trucks due to their limited payload and range, this probability ensures that drones are utilized effectively without overburdening them, while trucks handle larger and longer routes.

4.2 Individual repair and evaluation

In this paper, capacity constraints of both trucks and drones are addressed through repair operators before decoding individuals. For drones, any assignment violating payload limits or exceeding the minimum required energy is corrected by reassigning the corresponding customer to a truck. For trucks, if the total assigned demand surpasses capacity, a heuristic redistributes the excess load to the truck with the lowest workload, thereby enhancing feasibility and balancing resource utilization. Additionally, efficient drone operations require the selection of launch and landing locations due to limited battery capacity. Misaligned launches can increase waiting times or waste energy. To address this, we propose a time synchronization mechanism coordinating trucks, drones, and customer operations. As illustrated in Fig. 3, PFG-TS identifies feasible launch points at the depot and customer nodes, computes launch times based on order ready times and drone operations, and selects locations to minimize customer waiting and energy use. Corresponding landing points, either the depot or truck stops, are chosen to reduce travel to the next customer, while cumulative energy consumption is checked to ensure battery limits are not exceeded and total expected load on the truck does not exceed the capacity of the truck. Additionally, a consecutive sequence of customers is considered for drone service before landing, as long as the restricted energy constraint is satisfied. Algorithm 2 demonstrates detailed procedures for individual decoding to obtain the solution. If no suitable launching and landing points are found, the customer is reassigned to the corresponding truck in the individual's representation. Then, based on the obtained solutions, each individual is evaluated for fitness according to Eq. 6.

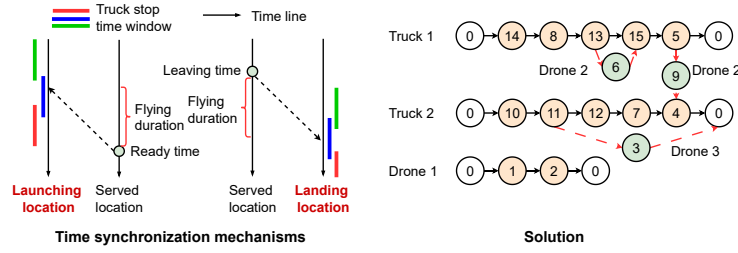


Fig. 3: Time synchronization mechanism and vehicles scheduling.

4.3 Reproduction

To conduct the Pareto front grid, the ideal and nadir points are first determined from the current population $\mathbf{P}^G$ with n_o objectives: $z_m^{*,G} = \min_{\mathbf{x}^G \in \mathbf{P}^G} f_m(\mathbf{x}^G)$ and $z_m^{nad,G} = \max_{\mathbf{x}^G \in \mathbf{P}^G} f_m(\mathbf{x}^G)$, $\forall m = 1, \dots, n_o$. The range of each objective m is then divided into GK uniform intervals, where the length of each segment is $d_m^G = (z_m^{nad,G} - z_m^{*,G} + 2\varepsilon)/GK$, with $\varepsilon > 0$ being a small constant. Each solution $\mathbf{x}^{i,G}$ is mapped into a grid coordinate based on its objective values: $Grid_m(\mathbf{x}^{i,G}) = \lceil (f_m(\mathbf{x}^{i,G}) - z_m^{*,G} + 2\varepsilon)/(d_m^G) \rceil$. Hence, the overall grid position of $\mathbf{x}^{i,G}$ is $Grid(\mathbf{x}^{i,G}) = (Grid_1(\mathbf{x}^{i,G}), \dots, Grid_{n_o}(\mathbf{x}^{i,G}))$. Individuals that fall into the same grid segment for a given objective m are grouped as $\mathbf{S}_{m,gk}^G = \{\mathbf{x}^G \mid Grid_m(\mathbf{x}^G) = gk\}$, $gk = 1, \dots, GK$. Then, we obtain all Pareto front grids, denoted as $PFG_m^G = \{PFG_{m,1}^G, \dots, PFG_{m,GK}^G\} \in PFG^G, \forall m = 1, \dots, n_o$, where $PFG_{m,gk}^G$ are non-dominated individuals in $\mathbf{S}_{m,gk}^G$.

Regarding parent selection, once the mating pool MP^G is formed, two parents are selected either from consecutive grids with probability ϵ , or from the entire population otherwise:

$$MP^G = \begin{cases} PFG_{m,gk}^G \cup PFG_{m,gk+1}^G, & \text{if } \mathcal{U}[0, 1] < \epsilon, \\ \mathbf{P}^G, & \text{otherwise,} \end{cases} \quad (12)$$

where $\mathcal{U}[0, 1]$ is a uniform random variable. For the crossover operator, at the upper layer, a partially matched crossover (PMX) is applied to preserve relative order and feasibility. Regarding the lower layer, representing the assignment of requests to trucks or drones, the binary value is exchanged according to the same crossover points. Mutation is applied with low probability to introduce additional diversity. Specifically, a swap mutation is used at the upper layer by exchanging two customer positions, while a bit-flip mutation is used at the lower layer to exchange the assignment of a customer between truck and drone. This strategy allows exploration of new solutions while maintaining feasible structures.

4.4 Selection and Terminated Condition

The population size remains constant across generations, allowing only well-adapted individuals to survive while less fit ones are eliminated, gradually improving solution quality. Selection is performed using the non-dominated sorting and crowding distance mechanisms of NSGA-II [22], and the algorithm terminates upon reaching the maximum number of generations.

Algorithm 2 Individual decode

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1:  $\mathcal{A} = \{A^k\}, \tilde{\mathcal{A}} = \{\tilde{A}^d\}, \forall k \in \mathcal{K}, d \in \mathcal{D} \leftarrow$  set of assigned customer based on individual representation;
2: Initialize  $\mathcal{S}_T \leftarrow \{\text{init\_truck\_solution}(A^k), \forall k \in \mathcal{K}\};$   $\triangleright$  Route of trucks
3:  $\mathcal{S}_D \leftarrow \emptyset;$   $\triangleright$  Route of drones
4: for  $d \in \mathcal{D}$  do
5:   Sort customers in  $A^d$  by ready time;
6:    $U \leftarrow$  set of unserved customers in  $A^d$ ;
7:    $T \leftarrow \emptyset;$   $\triangleright$  list of drone sub-trips
8:   while  $U \neq \emptyset$  do
9:      $u \leftarrow$  {first customer in  $U$ };
10:     $\mathcal{L} \leftarrow \text{findLaunch}(s, \mathcal{S}_T);$   $\triangleright$  From depot  $s$  or available points on truck routes
11:    Select  $l^* \in \mathcal{L}$ , which minimizes waiting time and energy;
12:     $\mathcal{H} \leftarrow \text{findLand}(s, \mathcal{S}_T);$   $\triangleright$  From depot  $s$  or available points on truck routes
13:    Select  $h^* \in \mathcal{H}$ ; which minimizes distance to next customer;
14:    if  $l^*$  or  $h^*$  does not exist then
15:      return  $u;$   $\triangleright$  Assign customers in  $u$  to truck
16:    end if
17:    Construct drone sub-trip  $\tau = (l^*, \text{customers}, h^*);$ 
18:    Update  $\tau$  and  $u$  if the drone serves some consecutive customers;
19:    update\_truck\_solution $(\mathcal{S}_T, \tau);$  Update  $U \leftarrow U \setminus \{u\}; T \leftarrow T \cup \{\tau\}$ 
20:  end while
21:   $\mathcal{S}_D \leftarrow \mathcal{S}_D \cup T$ 
22: end for
23: return  $(\mathcal{S}_T, \mathcal{S}_D)$ 

```

5 Experiments and Results

The dataset is based on the Solomon benchmark to ensure realism and diversity, containing 180 instances categorized by 100, 200, and 400 customers. Each instance follows the naming format `[dis][category]_[type]_[index]`, where `dis` denotes customer distribution (R: random, C: clustered, RC: combined), `category` indicates arrival time tightness (1: narrow, 2: wide), and `type` specifies the number of locations. The system parameters are adopted from [17]. We evaluate PFG-TS against other state-of-the-art multi-objective algorithms, including NSGA-II [23], MOEA/D [11], and NSGA-III [25]. All algorithms start from the same initial population, including 100 individuals, with a maximum of 100 generations, a crossover rate of 0.85, and a mutation rate of 0.1. Regarding the proposed algorithm, $GK = 5$ and $\varepsilon = 0.1$. Multi-objective performance is evaluated using HV and IGD [12]. To compute these metrics, the nadir point, which comprises the worst values of the objective functions across all obtained solutions, is used to normalize objective values, and the reference point $(1, 1, 1)$ is selected accordingly. Since the true Pareto front $\mathcal{F}^*$ is unknown, an approximate front is constructed by aggregating all non-dominated solutions found by the considered algorithms. Additionally, we assess the impact of various factors on algorithm performance, including the truck-assignment probability, the number of segments in the Pareto decomposition, and crossover operators.

Table 2: Comparison of HV and IGD results on different datasets

Dataset		HV				IGD			
No. Dis.		MOEA/D	NSGA-II	NSGA-III	PFG-TS	MOEA/D	NSGA-II	NSGA-III	PFG-TS
100	C	0.1422	0.3444	0.2694	0.4733	0.4250	0.1106	0.2117	4.8E-5
	R	0.2259	0.4649	0.3893	0.5444	0.3698	0.0823	0.1631	5.3E-5
	RC	0.1315	0.4020	0.3176	0.4759	0.4461	0.0864	0.1934	4.4E-3
200	C	0.0999	0.2339	0.1664	0.3483	0.3213	0.1246	0.2243	2.8E-4
	R	0.0924	0.2379	0.2092	0.3458	0.3334	0.1068	0.1589	5.1E-4
	RC	0.0930	0.2719	0.2560	0.3664	0.3622	0.0891	0.1331	~ 0
400	C	0.0657	0.1746	0.1290	0.2737	0.2864	0.1142	0.3371	~ 0
	R	0.0706	0.2137	0.1881	0.2692	0.2599	0.0590	0.0806	~ 0
	RC	0.0752	0.2160	0.2135	0.2859	0.2652	0.0725	0.0741	3.1E-4
Aver.		0.1107	0.2844	0.2376	0.3759	0.3410	0.0939	0.1751	6.2E-4

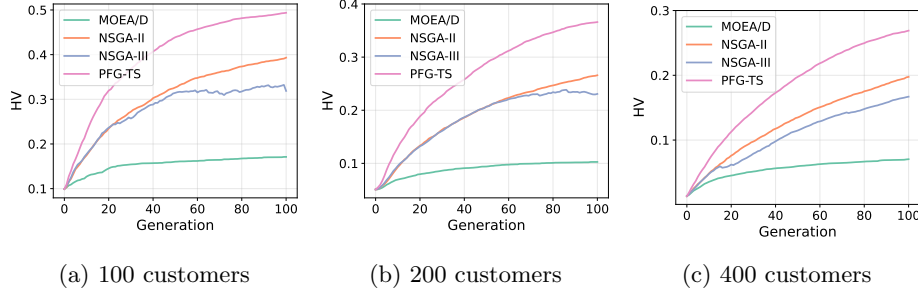


Fig. 4: HV convergence trend of PFG-TS compared to other algorithms.

Table 2 demonstrates the average HV and IGD values across different datasets. The results show that the proposed PFG-TS consistently achieves the best performance in most all cases, as highlighted in the table. For example, at the 200-customer clustered instance, PFG-TS improves HV by approximately 49% compared to NSGA-II and by 109% compared to NSGA-III. In terms of HV across all instances, the HV of PFG-TS is approximately 239.5%, 58.2%, and 32.2% higher than that of MOEA/D, NSGA-III, and NSGA-II, respectively. A similar trend can be observed in IGD, where PFG-TS consistently outperforms the compared algorithms. In some cases, the IGD values are close to 0, indicating that the majority of solutions on the approximate optimal Pareto front are contributed by PFG-TS. Additionally, Fig. 4 plots the convergence trends of the algorithms with the same initialized population. While compared methods such as MOEA/D and NSGA-III converge prematurely to local optima, and NSGA-II improves slowly after 100 generations, PFG-TS continues to make steady progress until convergence. This behavior highlights the ability of the posed algorithm to maintain exploration while refining solution quality.

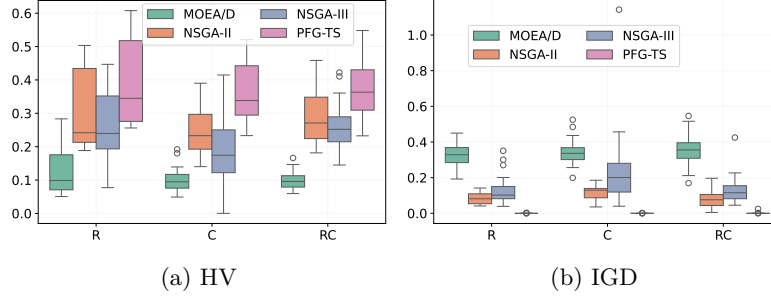


Fig. 5: Impact of customer distribution on the efficiency of algorithms.

Table 3: Impact of truck-assigned probability on the efficiency of PFG-TS

Data		HV						IGD					
α		0.5	0.6	0.7	0.8	0.9	1	0.5	0.6	0.7	0.8	0.9	1
Dis.	r	0.697	0.679	0.697	0.67	0.663	0.618	0.011	0.023	0.006	0.031	0.034	0.083
	c	0.566	0.605	0.573	0.54	0.529	0.522	0.043	0.009	0.034	0.063	0.062	0.072
	rc	0.599	0.592	0.601	0.588	0.584	0.563	0.023	0.020	0.019	0.025	0.030	0.047
Cat.	1	0.627	0.631	0.620	0.625	0.59	0.579	0.021	0.019	0.022	0.023	0.044	0.063
	2	0.614	0.62	0.627	0.574	0.595	0.556	0.033	0.014	0.020	0.056	0.036	0.070
Aver.		0.620	0.626	0.624	0.599	0.592	0.568	0.027	0.017	0.021	0.039	0.040	0.067

Fig. 5 investigates the impact of customer distribution on algorithmic efficiency in terms of HV and IGD. As shown in Fig. 5(a), PFG-TS consistently achieves higher HV values than the competing algorithms under all three scenarios. This indicates that PFG-TS is able to approximate the Pareto front more effectively regardless of the spatial distribution of customers. In particular, the improvement is more evident in the clustered case, where traditional algorithms such as MOEA/D and NSGA-III exhibit significant performance degradation. This can be explained by the fact that clustering the customer set allows more efficient collaboration between trucks and drones. In addition, Fig. 5(a) also illustrates the stability of the proposed algorithm in terms of HV values across different distributions, highlighting its generalization capability. A similar trend can be observed in Fig. 5(b) for IGD. The proposed PFG-TS attains the lowest IGD values across all distributions, confirming that its solutions are both closer to the approximate optimal Pareto front and more evenly spread. Notably, the robustness of PFG-TS is highlighted by its stable performance under the RC scenario, which is often considered the most challenging due to the combination of random and clustered customer placements.

Table 3 analyzes the impact of the truck-assignment probability α in the initialization step on the performance of PFG-TS. It is emphasized that $\alpha = 1$ corresponds to assigning all customers to trucks, thereby eliminating the role of drones at the beginning. The results show that such settings significantly degrade the solution quality, as both HV and IGD are consistently worse when $\alpha = 1$ compared to other values. In contrast, moderate probabilities yield superior outcomes. Specifically, $\alpha = 0.6$ achieves the highest average HV (0.626) and

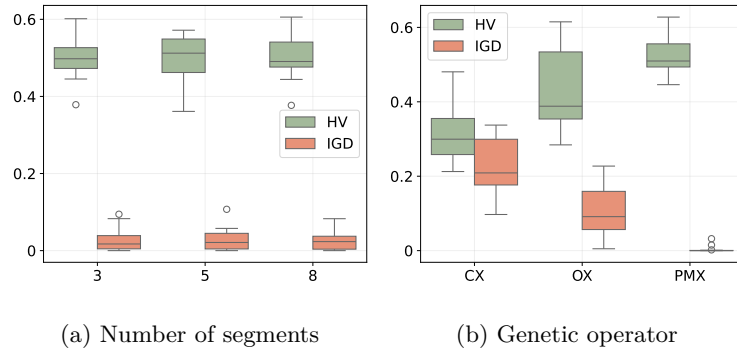


Fig. 6: Impact of number of segments and genetic operators on PFG-TS.

the lowest average IGD (0.017), representing an improvement of about 10.21% in HV and a 74.63% reduction in IGD compared to $\alpha = 1$. Furthermore, for each type of data, there are specific values of α that yield the best performance, with a focus on 0.6 and 0.7. This can be explained by the fact that a small α results in more tasks being assigned to drones, whereas a large α fails to fully exploit the robustness of the drones. These findings highlight that balancing the initial allocation between trucks and drones facilitates more effective search exploration, thereby enhancing the robustness of the proposed approach.

In Fig. 6(b), we compare three different genetic operators, including Cycle Crossover (CX), Order Crossover (OX), and Partially-Mapped Crossover (PMX) [2]. The results indicate that PMX achieves the best overall performance, with consistently higher HV and lower IGD values, while CX and OX are less effective. This is because PMX is particularly suitable for permutation encoding, as it preserves the relative order and position of elements, maintaining valid offspring and reducing disruption of good sub-tours in routing problems. These findings demonstrate that the choice of crossover operator has a significant impact on the search efficiency of PFG-TS, particularly in effectively exploring the solution space for time-synchronized truck-drone routing problem.

6 Conclusions

This paper introduces the ITDDPDE problem, jointly scheduling trucks and drones while considering payload-dependent drone endurance and flexible launch and retrieval points. We propose PFG-TS, a Pareto-based evolutionary algorithm with two-level encoding and a heuristic for drone operations. Numerical experiments show that PFG-TS outperforms state-of-the-art multi-objective algorithms in terms of HV and IGD, demonstrating the efficiency of integrated truck-drone systems. Future work can explore heterogeneous fleets and dynamic routing to further enhance practical applicability.

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